The last two chapters round out the book with a focus on administration and performance:

- Chapter 14 examines the administrative side of productionizing Elasticsearch. This includes understanding how to scale the cluster under various loads, communication between nodes, and shard sizing. The chapter also explores the crucial concept of snapshotting, providing practical examples of taking a snapshot and retrieving data from it when needed. Advanced configurations and the cluster master concept are also thoroughly examined.
- Chapter 15 dives into troubleshooting a poorly performing or issue-ridden Elasticsearch cluster. The chapter examines common causes, such as search and speed bottlenecks, unstable and unhealthy clusters, and circuit breakers, among others. This chapter will equip you with the knowledge to diagnose and address performance issues, ensuring that your Elasticsearch cluster runs smoothly and efficiently.

The book also has three appendixes:

- Appendix A is a practical guide to installing Elasticsearch and Kibana in your local environment.
- Appendix B examines ingest pipelines, a key component of data preprocessing in Elasticsearch, and how to configure and use them in various scenarios.
- Appendix C covers Elasticsearch's interoperability with various programming languages through clients such as Java, Python, JavaScript, C#, and others, offering examples and best practices.

About the code

One of the primary goals of this book is to provide a seamless hands-on experience by including easily executable code. After several iterations, a decision was made to host all the queries, written and executed on Kibana, in a text file on GitHub. These queries are captured as Query DSL-based JSON code. The aim is to provide a straightforward process where you can copy the text file from GitHub and paste it into your Kibana Dev Tools application for immediate execution.

To further aid in your learning journey, sample data files and, when necessary, mappings for these indexes are provided in a dedicated datasets folder. This approach ensures a practical, learner-friendly experience that allows you to engage directly with the material and apply your newfound knowledge.

The source code is available on GitHub (https://github.com/madhusudhankonda/elasticsearch-in-action) and the book's website (https://www.manning.com/books/elasticsearch-in-action-second-edition). The folders are as follows:

- *kibana_scripts*—Query DSL scripts for each chapter.
- *datasets*—Mappings and sample data sets required for the book chapters.
- *code*—Java and Python code.